

# FinTwit Prediction Audit

An LLM-Powered Statistical Audit of ~18,000 Stock Predictions  
by 51 Financial Influencers on X (2021–2025)

*Aggregate directional accuracy 45.0%; no account shows durable skill across market regimes*

# 1. Introduction

Financial social media — 'FinTwit' — hosts a large and vocal community of traders and investors who publish directional stock predictions publicly and in real time. These accounts attract substantial followings and, implicitly, influence retail investment behaviour. Yet there exists almost no rigorous, large-scale audit of whether these predictions have any actual merit. This project fills that gap.

Using approximately 18,000 tweets from 51 financial-influencer accounts on X (formerly Twitter), spanning January 2021 to February 2025, we conduct a reproducible statistical audit that answers three pre-registered research questions:

## 1.1 Research Questions

RQ1 — Accuracy & benchmark. How accurate are FinTwit predictions directionally, and do they beat a passive market benchmark (buy-and-hold of the named stock, and the broad market via SPY) over the matched prediction horizon?

RQ2 — Skill vs luck. Across all 51 accounts, after correcting for multiple comparisons, is any individual account's track record statistically distinguishable from chance? Model true per-account skill with an empirical-Bayes / hierarchical shrinkage estimator that accounts for small-sample noise.

RQ3 — Signal value. Do tweet-level signals — sentiment, trade type, horizon, engagement, and the LLM's own self-reported confidence — carry any information about whether a prediction comes true? Includes (a) a formal calibration analysis of the LLM confidence score (a preliminary check already shows it is essentially uncalibrated,  $r \approx 0$ ), and (b) a predictive model of `prediction_correct`.

## 1.2 Provenance

This is an anonymised personal rebuild of a former paid consulting engagement, presented as an independent portfolio project. No original client is named or referenced anywhere in this analysis.

# 2. Data

## 2.1 Overview

The dataset comprises 18,071 tweets posted by 51 financial-influencer accounts on X between 2021-01-07 and 2025-02-20. The raw tweet text and all platform metadata were collected prior to this analysis; no Twitter/X API access is required to reproduce the results. All 51 account handles have been pseudonymised to `account_01` ... `account_51`,

assigned deterministically by alphabetical sort order of the original handles. The handle-to-pseudonym mapping is stored in `data/account_mapping.csv`, which is excluded from version control.

## 2.2 Derivation of `tweets_raw.csv`

The working dataset `data/tweets_raw.csv` was derived from the source export by (i) retaining only genuinely raw columns (tweet identifiers, author, tweet text, timestamps, reply structure, engagement counts, and account metadata), (ii) dropping all LLM-derived and validation columns that will be regenerated in later phases (tickers\_mentioned, time\_horizon, trade\_type, sentiment, confidence, price\_change\_pct, prediction\_correct, actual\_return, and related fields), (iii) parsing types (tz-aware datetimes, nullable integers, booleans), and (iv) replacing all real account handles with their pseudonyms.

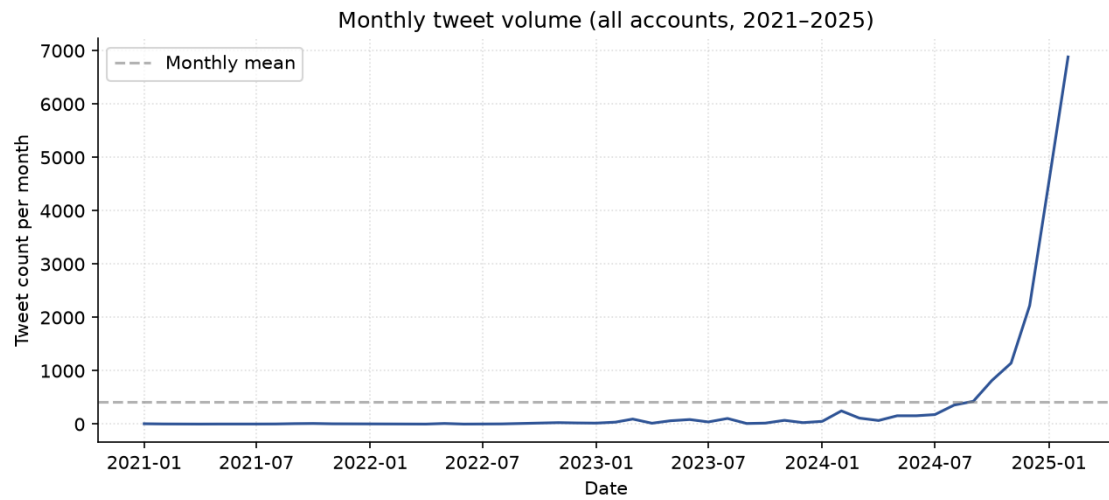
## 2.3 Columns

`tweets_raw.csv` contains 17 columns. Full definitions, dtypes, and notes are provided in `data/DATA_DICTIONARY.md`. In brief: `tweet_id`, `conversation_id`, and `reply_to_tweet_id` are opaque string identifiers; `tweet_type` records the thread role (parent or reply); `author` and `reply_to_user` are pseudonymised handles; `text` is the raw tweet body and the sole NLP input for all subsequent phases; `created_at` is a tz-aware UTC timestamp and `created_date` is the corresponding calendar date; `likes`, `retweets`, `replies_count`, and `views` are nullable integer engagement counts; `author_followers`, `author_following` are nullable integer account-size metrics; `author_verified` and `author_blue_verified` are booleans.

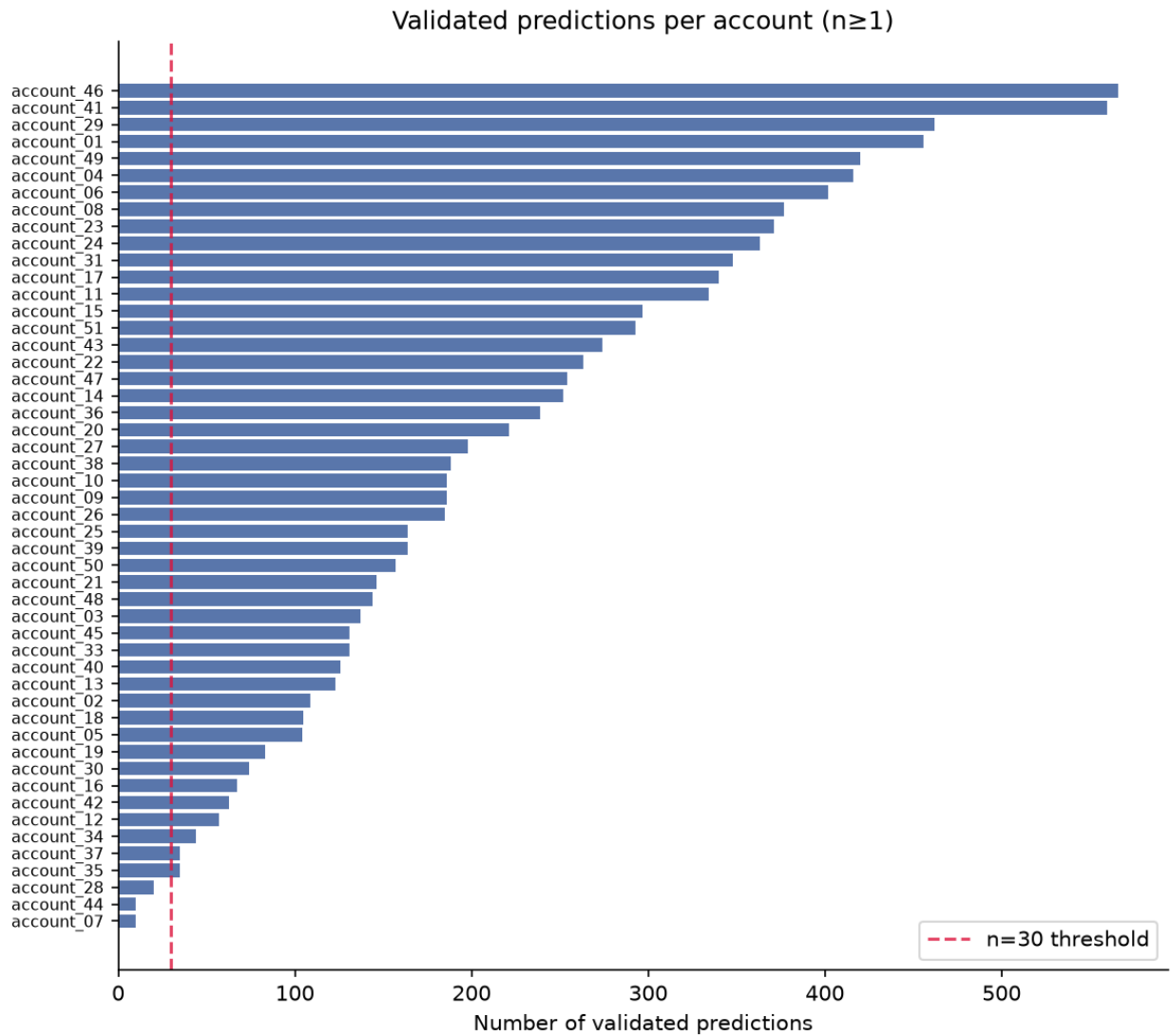
## 2.4 Data-quality caveats

**ID precision:** the source export stored `tweet_id`, `conversation_id`, and `reply_to_tweet_id` as float64 (scientific notation, e.g. `1.8924e+18`). Float representation of 64-bit tweet IDs loses the least-significant digits, so these identifiers are treated as opaque labels only and are never used as exact join keys against external datasets.

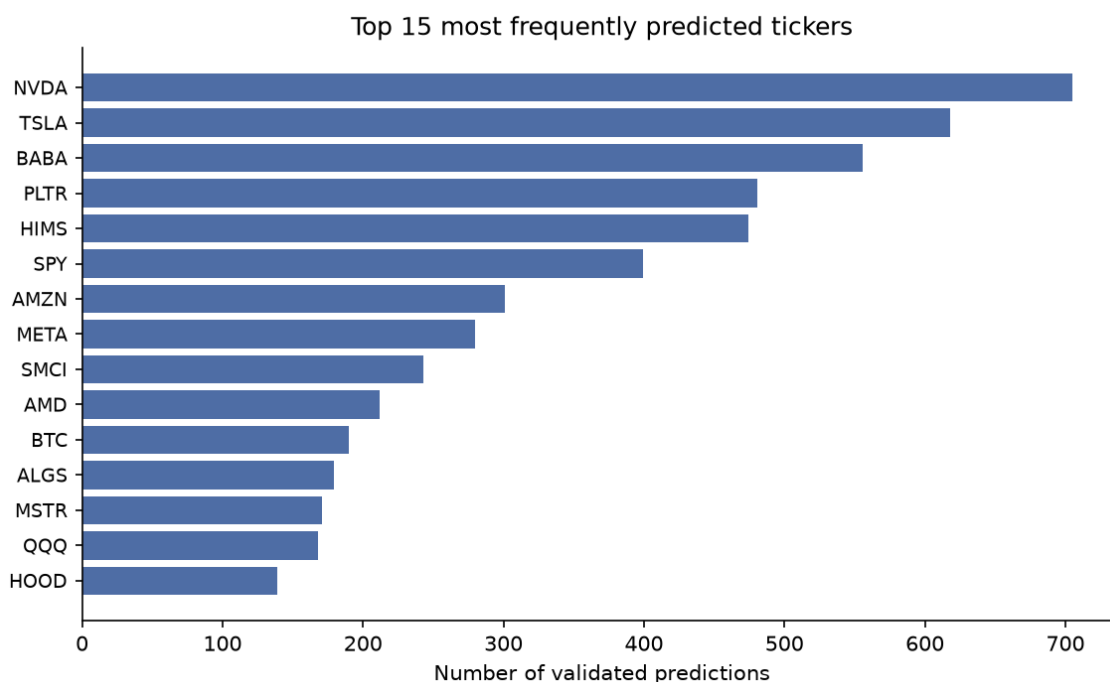
**Missing engagement data:** engagement and account-size columns contain missing values, particularly for older tweets and reply-type tweets. Nullable Int64 dtype is used throughout to preserve the distinction between a genuine zero and a missing observation.



**Figure 2.1.** *Monthly tweet volume across all 51 accounts (2021–2025). Dashed line shows the overall monthly mean.*



**Figure 2.2.** Number of validated predictions per account (all 51 accounts). Dashed red line marks the  $n=30$  minimum threshold used in RQ2.



*Figure 2.3. Top 15 most frequently predicted tickers in the validated dataset.*

### 3. Ticker Extraction

Before any directional prediction can be validated against market data, we must identify which stock(s) each tweet is referring to. We use a two-stage pipeline that prioritises speed and cost efficiency: a regex baseline captures the vast majority of cases, and an LLM fallback handles the remainder.

#### 3.1 Regex Baseline

The primary extraction method is a regular expression that matches cashtag patterns of the form \$TICKER, where TICKER is between one and five ASCII letters. The pattern is case-insensitive and requires that no additional letters immediately follow the match (word boundary), ensuring that phrases like '\$AAPLisgreat' are not falsely matched.

A stoplist removes tokens that match the pattern but are not stock tickers: fiat-currency abbreviations (USD, EUR, GBP, JPY, CAD, AUD, CHF, CNY, and others), and common financial abbreviations that appear as cashtags (IPO, ETF, CEO, AI, etc.). Dollar-prefixed numbers such as \$100 are excluded automatically because the pattern requires letters. All surviving matches are upper-cased and de-duplicated within each tweet.

## 3.2 LLM Fallback

The LLM fallback is invoked only for tweets in which the regex finds no cashtag — roughly 27.6% of the corpus. This design keeps API costs proportional to the hardest cases rather than the full dataset. The fallback uses gpt-4o-mini with a zero-temperature structured-JSON prompt that instructs the model to resolve unambiguous company-name mentions to their primary US ticker (e.g. 'Tesla' to TSLA), return [] when no specific stock is discussed, and never invent a ticker it is uncertain about.

Results are cached on disk (cache/llm\_tickers/) keyed by the SHA-256 hash of the tweet text. A re-run on the same data incurs zero additional API cost. The OpenAI API key is read from the OPENAI\_API\_KEY environment variable; extraction degrades gracefully to regex-only when the key is absent. Rate-limit and transient API errors are retried with exponential back-off (up to four attempts).

## 3.3 Output Columns

The extraction step adds two columns to the working dataset (data/tweets\_with\_tickers.csv): tickers\_mentioned, a Python list of deduplicated, upper-cased ticker strings (may be empty); and has\_ticker, a boolean that is True if and only if tickers\_mentioned is non-empty. No filtering for tradeable / exchange-listed status is applied at this stage — that validation occurs in Phase 3 via yfinance.

## 3.4 Coverage

Running the full two-stage pipeline over all 18,071 tweets produced the following coverage:

Regex hit: 13,089 tweets (72.4%). LLM-only hit: 950 tweets (5.3%). No ticker found: 4,032 tweets (22.3%). At least one ticker (regex + LLM combined): 14,039 tweets (77.7%).

The LLM fallback added 950 tweets that the regex missed — recovering 5.3 percentage points of coverage by resolving company-name mentions that lacked a cashtag (e.g. 'Tesla' to TSLA). Roughly 4,032 tweets (22.3%) yielded no ticker from either method, consistent with the expectation that a share of the corpus consists of market commentary, retweets, or general observations not tied to a specific stock. All 4,982 no-cashtag tweets were sent through the LLM; results are cached on disk (cache/llm\_tickers/, 4,705 cache files) so re-runs make zero additional API calls.

## 3.5 Evaluation Methodology and Results

To assess extraction quality, we drew a stratified random sample of 100 tweets (random seed 42) covering both cashtag-containing and non-cashtag tweets. Each tweet's text was read and the correct tickers entered by hand, blind to the automated outputs stored in `data/ticker_eval_sample.csv`. Once labelled, `scripts/eval_tickers.py` computed precision, recall, and F1 at the ticker level, separately for the regex-only and combined (regex + LLM) methods.

Table 3.1 — Ticker extraction performance (n = 100 hand-labelled tweets)

| Method      | TP  | FP | FN | Precision | Recall | F1    |
|-------------|-----|----|----|-----------|--------|-------|
| Regex only  | 114 | 3  | 8  | 0.974     | 0.934  | 0.954 |
| Regex + LLM | 117 | 4  | 5  | 0.967     | 0.959  | 0.963 |

The regex baseline achieves high precision (0.974) — virtually every cashtag it extracts is a genuine ticker — but misses 8 tickers that appear as company-name mentions without a cashtag (recall 0.934). Adding the LLM fallback recovers 3 of those 8 misses, raising recall to 0.959 at the cost of just one additional false positive (FP 3 → 4), yielding a net F1 improvement from 0.954 to 0.963. Both methods perform well; the LLM contributes a modest but consistent recall gain for company-name mentions.

## 4. Classification Methodology

Each of the 14,039 ticker-bearing tweets was classified by `gpt-4o-mini` into four dimensions that define the nature of the prediction. Tweets without a ticker (4,032 rows) are left unclassified and receive null values for all classification columns.

### 4.1 Classification Schema

The model was asked to return a strict JSON object with exactly four fields:

`sentiment` — the tweet's directional view on the named stock(s): "bullish" (positive or long bias), "bearish" (negative or short bias), or "neutral" (no clear directional view).

`time_horizon` — the implied prediction window: "short\_term" (days to approximately one month), "medium\_term" (approximately one to six months), "long\_term" (more than six months), or "unknown" when no horizon is implied.

`trade_type` — the nature of the tweet: "trade\_suggestion" (an explicit buy, sell, entry, or exit recommendation), "analysis" (a reasoned opinion or chart reading without an explicit trade call), "news" (relaying factual information such as earnings or macro data), or "general\_discussion" (everything else).



confidence — an integer from 0 to 100 representing how assertive or certain the tweet sounds, where 0 is very tentative and 100 is extremely certain. This score reflects the linguistic register of the tweet, not a calibrated probability of correctness; its calibration against actual prediction outcomes is assessed separately in Phase 6 (RQ3).

## 4.2 Implementation and Caching

Classification calls used gpt-4o-mini at temperature=0 with response\_format=json\_object to enforce structured output. The API key is read from the OPENAI\_API\_KEY environment variable. Rate-limit and transient API errors are retried with exponential back-off (up to four attempts). On any failure — including malformed or truncated JSON — the row falls back to safe defaults (sentiment: neutral, time\_horizon: unknown, trade\_type: general\_discussion, confidence: 0) and the event is logged; the pipeline never crashes mid-run.

Results are cached on disk (cache/llm\_classify/) keyed by the SHA-256 hash of the tweet text. Re-running the classification script on the same corpus makes zero additional API calls; all 14,039 results are served from the cache.

## 4.3 Results and Distributions

All 14,039 ticker-bearing tweets were classified successfully. 13,730 calls were made to the API; 309 were served from the on-disk cache (duplicate tweet texts). The resulting distributions are as follows.

Sentiment: bullish 8,847 (63.0%), bearish 2,951 (21.0%), neutral 2,241 (16.0%). The strong bullish skew is consistent with the promotional nature of financial-influencer content on X.

Time horizon: short\_term 6,718 (47.9%), unknown 3,793 (27.0%), medium\_term 2,105 (15.0%), long\_term 1,423 (10.1%). Nearly half of all predictions are framed as short-term calls, and more than a quarter carry no explicit time frame.

Trade type: analysis 6,307 (44.9%), trade\_suggestion 3,263 (23.2%), general\_discussion 2,899 (20.6%), news 1,570 (11.2%). The largest category is reasoned opinion without an explicit trade call; explicit buy/sell recommendations account for roughly one in four classified tweets.

Confidence scores across all 14,039 classified tweets: mean 74.2, median 75, min 10, max 100. The high median confidence reflects the assertive register typical of FinTwit content. Whether this expressed confidence bears any relationship to actual prediction accuracy is assessed in Phase 6 (RQ3 calibration analysis); preliminary findings suggest it does not.

## 4.4 Worked Example

Table 4.1 shows three representative tweets selected to illustrate the classification schema and downstream outcomes. Each example contains no account handles (anonymisation preserved). Tweet text is truncated to approximately 90 characters; all @-mentions are redacted to @user.

**Table 4.1. Classification worked example (three real tweets, text redacted or truncated).**

| Tweet text (~90 chars)                                                                                                           | Ticker | Sentiment | Horizon     | Trade type       | Conf. | Outcome |
|----------------------------------------------------------------------------------------------------------------------------------|--------|-----------|-------------|------------------|-------|---------|
| \$NVDA to test 110–115                                                                                                           | NVDA   | bullish   | short_term  | trade_suggestion | 75    | Correct |
| Someone bought \$700,000 worth of \$TSLA 305 puts that expire next week — calling for a test of \$300                            | TSLA   | bearish   | short_term  | analysis         | 75    | Correct |
| Starting to think AI lending and insurance might be a really nice trade in the coming months. Everyone hates legacy insurance... | LMND   | bullish   | medium_term | trade_suggestion | 75    | Wrong   |

## 5. Market Validation

Phase 3 validates each directional prediction — those classified as bullish or bearish with at least one resolvable ticker — against actual stock-price data fetched from Yahoo Finance. Two binary outcomes are computed for every validated prediction: whether the prediction was directionally correct (stock moved in the predicted direction), and whether it beat the passive market benchmark (the named stock out- or underperformed SPY over the same window).

### 5.1 Validation Methodology

For each qualifying prediction the following steps are applied. First, the entry date is set to the first trading day on or after the tweet's created\_date, ensuring we do not assume a trade executed before the tweet was public. Second, the target date is calculated as the entry date plus the horizon window in trading days (see §5.2). Third, look-ahead safety is enforced: predictions whose target date has not yet passed as of the run date are excluded rather than filled with partial data. Fourth, the adjusted-close prices of the stock and SPY at the entry and target dates are retrieved; 'adjusted close' accounts for stock splits and dividend distributions.

A prediction is judged correct if the stock's return over the window matches the predicted direction: positive return for bullish calls, negative return for bearish calls. A flat return (exactly zero) is treated as incorrect. A prediction beats the market if its excess return (stock return minus SPY return over the same window) is signed in the predicted direction: positive excess for bullish, negative excess for bearish. These two outcomes are independent — a bullish call on a stock that rises 2% when SPY rises 5% is directionally correct but does not beat the market.

Only the first ticker in `tickers_mentioned` is used for validation. This first-ticker rule is a deliberate simplification: when a tweet names multiple stocks it is not clear which one the prediction concerns most strongly. Using the first-mentioned ticker is documented and applied consistently; sensitivity to this choice is discussed in §5.4.

## 5.2 Prediction Horizon Windows

The `time_horizon` label assigned in Phase 2 maps to a fixed number of trading days:

`short_term` → 21 trading days (approximately one calendar month). `medium_term` → 63 trading days (approximately one calendar quarter). `long_term` → 126 trading days (approximately six calendar months). `unknown` → 21 trading days (same as `short_term`).

Tweets assigned `horizon = 'unknown'` by the classifier are validated using the 21-day window and flagged with `horizon_was_unknown = True` in the output dataset. This allows downstream analyses to exclude or stratify these rows if the arbitrary horizon assignment is a concern. The 21-day default is chosen because `short_term` is the modal horizon in the corpus (47.9% of classified tweets), and financial influencer content is disproportionately oriented toward short-term calls.

## 5.3 Price Data and Caching

Price histories are fetched from Yahoo Finance via the `yfinance` library (adjusted close, daily frequency, from 2020-01-01 to the run date). Each ticker's full history is downloaded once and cached to `cache/prices/{TICKER}.csv`. Subsequent runs read entirely from disk, making re-runs instantaneous and reproducible without internet access. Tickers that return an empty history (delisted symbols with no data, crypto tickers without the exchange suffix, etc.) receive a `.notfound` sentinel file so they are never retried. SPY is fetched and cached using the same mechanism.

## 5.4 Validation Funnel

Applying the validation pipeline to all 18,071 tweets yields the following funnel. Of 11,798 directional predictions (`has_ticker` AND sentiment in `{bullish, bearish}`), 1,102 were excluded because the first ticker returned an empty price history from Yahoo

Finance — delisted stocks, crypto tokens without an exchange suffix (e.g. DOGE, SHIB, SOL), indices not available as Yahoo Finance tickers (e.g. SPX, VIX, DXY), and non-US symbols without US ADR listings (e.g. ADYEN, LVMH, TSMC). A further 4 predictions had a target date that had not yet passed as of the run date and 2 had no matching entry date in the price series. This leaves 10,690 validated predictions (90.6% of the directional set).

Of the 10,690 validated predictions, 1,836 (17.2%) had `time_horizon = 'unknown'` and were assigned the 21-day window by default; these are flagged via `horizon_was_unknown = True` in `tweets_validated.csv` and can be excluded in downstream analyses if sensitivity to the default assignment is a concern.

Table 5.1. Data validation funnel from raw corpus to analysable predictions.

| Stage                            | Count  | % of prior stage | Notes                                |
|----------------------------------|--------|------------------|--------------------------------------|
| All tweets in corpus             | 18,071 | —                |                                      |
| Has at least one ticker          | 14,039 | 77.7%            | Regex + LLM extraction               |
| Directional (bullish or bearish) | 11,798 | 84.0%            | Neutral excluded                     |
| Price data found (validated)     | 10,690 | 90.6%            | Excl. 1,102 delisted/crypto; 6 other |

## 5.5 Prediction Accuracy

Across all 10,690 validated predictions, 4,813 were directionally correct (45.0%) and 4,744 beat the SPY benchmark (44.4%). Both figures are materially below the 50% threshold expected from a random coin flip, suggesting that FinTwit predictions on average destroy value relative to passive buy-and-hold. A formal statistical test of significance is conducted in Phase 4 (RQ1).

By sentiment: bullish calls ( $n = 8,083$ ) were correct 40.5% of the time and beat the market 40.5% of the time. Bearish calls ( $n = 2,607$ ) were correct 59.1% of the time and beat the market 56.3% of the time. The higher accuracy of bearish calls is consistent with the time period covered (2021-2025), which included the 2022 bear market during which many bearish predictions on individual stocks were vindicated. The bull-dominant sentiment distribution combined with sub-50% bullish accuracy largely explains the below-chance aggregate rate.

By prediction horizon: `short_term` ( $n = 5,774$ ) correct 41.6%; `medium_term` ( $n = 1,869$ ) correct 45.3%; `long_term` ( $n = 1,211$ ) correct 62.3%; `unknown/defaulted-to-21d` ( $n = 1,836$ ) correct 44.2%. The `long_term` outperformance reflects the strong upward trend in US equities over the five-year study window: a bullish long-term call on most large-cap US stocks made at any point in 2021-2025 was more likely to be correct purely due to the

overall market direction. This effect will be quantified and controlled for in the Phase 5 per-account analysis.

## 6. RQ1 — Accuracy and Market Benchmark

This section answers RQ1: how accurate are FinTwit predictions directionally, and do they beat a passive market benchmark? We assess two binary outcomes across all 10,690 validated predictions: `prediction_correct` (did the stock move in the predicted direction?) and `beats_market` (did the stock out- or underperform SPY over the same horizon?). We report 95% Wilson score confidence intervals throughout and test significance against a 50% null (a coin flip) using a two-tailed proportion z-test.

### 6.1 Overall Accuracy

Across all 10,690 validated predictions, 4,813 were directionally correct (45.0%; 95% CI [44.1%, 46.0%]). Against the market benchmark, 4,744 predictions beat SPY (44.4%; 95% CI [43.4%, 45.3%]). Both figures are statistically and practically significantly below the 50% benchmark expected from an uninformative coin flip.

Directional correct:  $z = -10.29$ ,  $p = 7.75e-25$ . Beats market:  $z = -11.63$ ,  $p = 3.05e-31$ . At a sample size of 10,690, these z-statistics are not merely crossing an arbitrary significance threshold — they reflect a systematic directional bias toward overconfident predictions. The aggregate portfolio of FinTwit calls actively destroys value relative to a passive buy-and-hold strategy in both the named stocks and the broad market.

Mean returns over the validated prediction windows: mean stock return -1.85% per prediction, mean SPY return -1.60%, mean excess return -0.24%. The negative mean stock and SPY returns reflect the 2022 market drawdown and the disproportionate short-term prediction horizon in the corpus (21 trading days, covered extensively during a high-volatility period). The negative mean excess return confirms the below-benchmark result on a magnitude basis, not just a directional one.

### 6.2 Why `beats_market` Is the Cleaner Skill Measure

Raw directional accuracy (`prediction_correct`) conflates genuine analytical skill with the unconditional tendency of equity markets to rise over time. A bullish call on any diversified portfolio of stocks held for 126 trading days (long\_term horizon) will be correct simply because markets exhibit a positive long-run drift. The `beats_market` metric removes this base-rate advantage: it asks whether the prediction captured incremental information beyond what a passive SPY position would have produced. This distinction matters most for long-term predictions and is examined in §6.3.

### 6.3 Accuracy by Segment

Table 6.1 presents accuracy and beats\_market rates with 95% Wilson CIs for each subgroup. Figure 6.1 shows the same results as a grouped bar chart with error bars. All subgroup estimates are below or at 50% except long\_term prediction\_correct (62.3%), which is explained below.

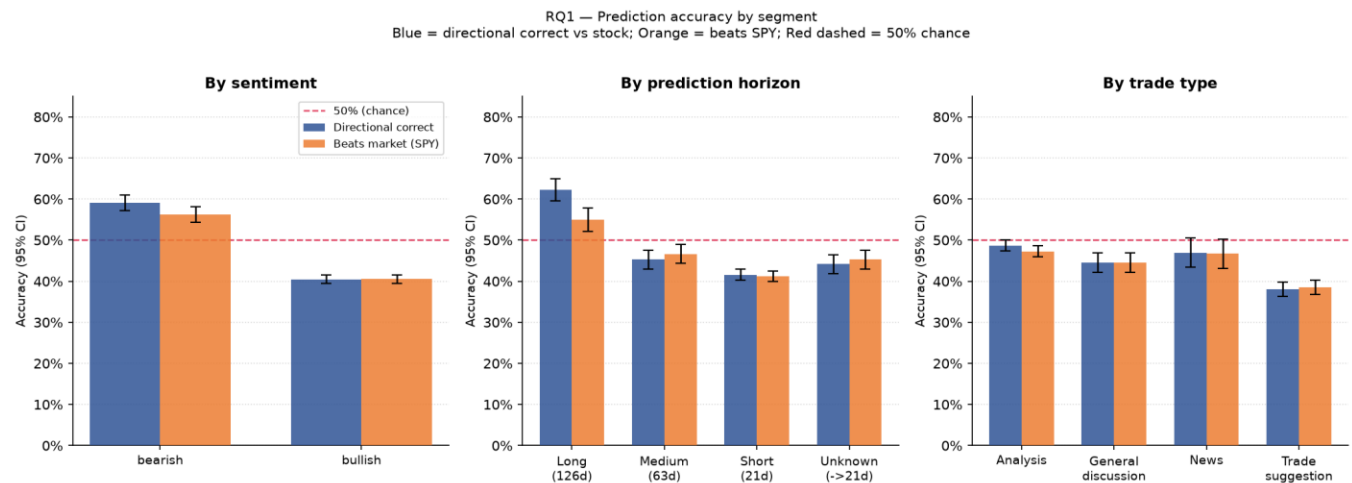
By sentiment — Bearish calls (n = 2,607) were correct 59.1% [57.2%, 60.9%] of the time and beat the market 56.3% [54.4%, 58.2%] of the time. Bullish calls (n = 8,083) were correct only 40.5% [39.4%, 41.6%] and beat the market 40.5% [39.5%, 41.6%]. The strong bearish outperformance on both metrics reflects the composition of the study period: the 2022 bear market vindicated many bearish predictions. Because bearish calls are a minority (22.1% of directional predictions), the overall aggregate rate is dragged down by the high-volume bullish calls, which performed substantially below chance. This is consistent with the promotional, hype-driven character of FinTwit content.

By prediction horizon — short\_term: 41.6% correct, 41.1% beats (n = 5,774); medium\_term: 45.3% correct, 46.7% beats (n = 1,869); long\_term: 62.3% correct, 55.0% beats (n = 1,211); unknown (defaulted to 21d): 44.2% correct, 45.3% beats (n = 1,836). The long\_term raw accuracy (62.3%) initially appears impressive but is substantially explained by the positive market drift over the 2021-2025 study window. The beats\_market rate for long\_term predictions drops to 55.0% — above 50%, but much closer to chance, and not dramatically different from the medium\_term beats\_market rate of 46.7%. Short-term predictions — the plurality of the corpus — perform worst on both metrics, consistent with short-term price moves being nearly impossible to predict.

By trade type — analysis: 48.7% correct, 47.2% beats (n = 5,333); news: 47.0% correct, 46.7% beats (n = 758); general\_discussion: 44.5% correct, 44.5% beats (n = 1,691); trade\_suggestion: 38.0% correct, 38.5% beats (n = 2,908). The most striking finding here is that explicit trade recommendations — the subset of tweets that make a specific, actionable buy or sell call — are the worst performers by a considerable margin, 12 percentage points below the analysis category on both metrics. Analysis-type tweets (chart readings, reasoned opinion pieces) perform closest to 50% but still below it. The ordering is: analysis > news > general\_discussion > trade\_suggestion, exactly the reverse of what skill would predict.

**Table 6.1. Accuracy and beats\_market rate by segment (95% Wilson CIs). \* p<0.05, \*\* p<0.01, \*\*\* p<0.001 vs 50% null (not shown here; see §6.1).**

| Group                          | n      | Correct % | 95% CI (correct) | Beats % | 95% CI (beats) |
|--------------------------------|--------|-----------|------------------|---------|----------------|
| Overall (all validated)        | 10,690 | 45.0%     | [44.1%, 46.0%]   | 44.4%   | [43.4%, 45.3%] |
| Sentiment: bearish             | 2,607  | 59.1%     | [57.2%, 60.9%]   | 56.3%   | [54.4%, 58.2%] |
| Sentiment: bullish             | 8,083  | 40.5%     | [39.4%, 41.6%]   | 40.5%   | [39.5%, 41.6%] |
| Horizon: short_term            | 5,774  | 41.6%     | [40.3%, 42.9%]   | 41.1%   | [39.9%, 42.4%] |
| Horizon: medium_term           | 1,869  | 45.3%     | [43.0%, 47.5%]   | 46.7%   | [44.4%, 48.9%] |
| Horizon: long_term             | 1,211  | 62.3%     | [59.5%, 64.9%]   | 55.0%   | [52.2%, 57.8%] |
| Horizon: unknown               | 1,836  | 44.2%     | [41.9%, 46.5%]   | 45.3%   | [43.0%, 47.5%] |
| Trade type: analysis           | 5,333  | 48.7%     | [47.4%, 50.1%]   | 47.2%   | [45.9%, 48.6%] |
| Trade type: news               | 758    | 47.0%     | [43.4%, 50.5%]   | 46.7%   | [43.2%, 50.3%] |
| Trade type: general_discussion | 1,691  | 44.5%     | [42.2%, 46.9%]   | 44.5%   | [42.1%, 46.8%] |
| Trade type: trade_suggestion   | 2,908  | 38.0%     | [36.3%, 39.8%]   | 38.5%   | [36.7%, 40.3%] |

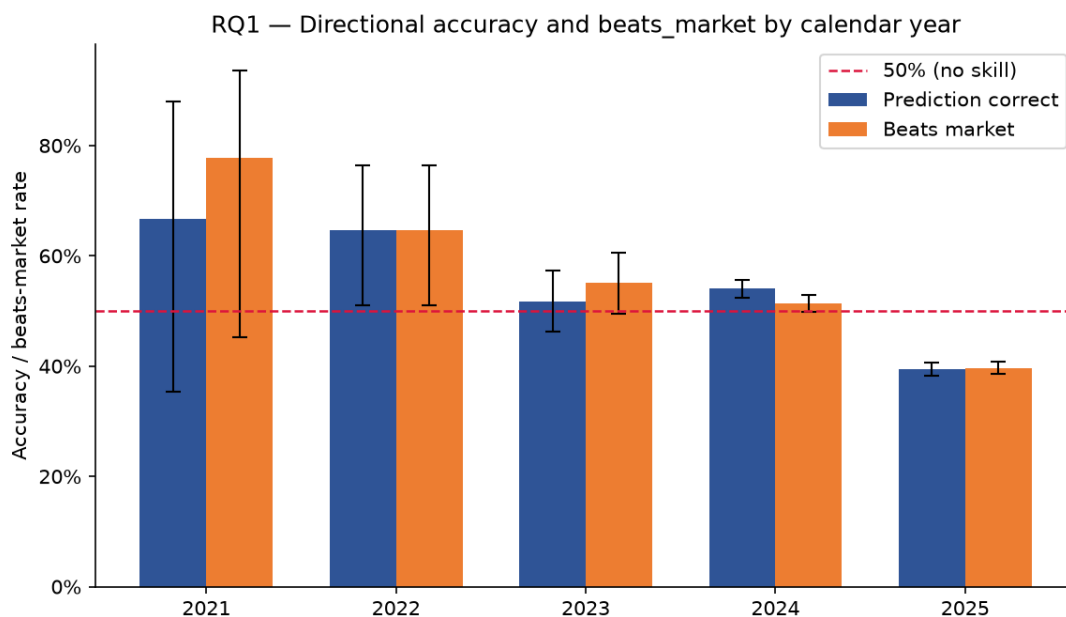


**Figure 6.1. Prediction accuracy and beats\_market rate by segment (sentiment, time horizon, trade type).**  
Error bars are 95% Wilson confidence intervals. Dashed line marks 50% (no-skill baseline).

## 6.4 Sensitivity: Unknown-Horizon Rows

Of the 10,690 validated predictions, 1,836 (17.2%) had time\_horizon = 'unknown' and were assigned the 21-day window by default (flagged as horizon\_was\_unknown). Excluding these rows yields n = 8,854 predictions. Results are virtually unchanged: directional correct 45.2% [44.2%, 46.2%] ( $p = 1.66e-19$ ); beats market 44.2% [43.2%, 45.2%] ( $p = 8.75e-28$ ). The headline finding is robust to the choice of a 21-day default for unspecified horizons.

## 6.5 Accuracy by Calendar Year



**Figure 6.2.** Directional accuracy and beats\_market rate by calendar year (2021–2025), with 95% Wilson confidence intervals. Dashed line marks 50%.

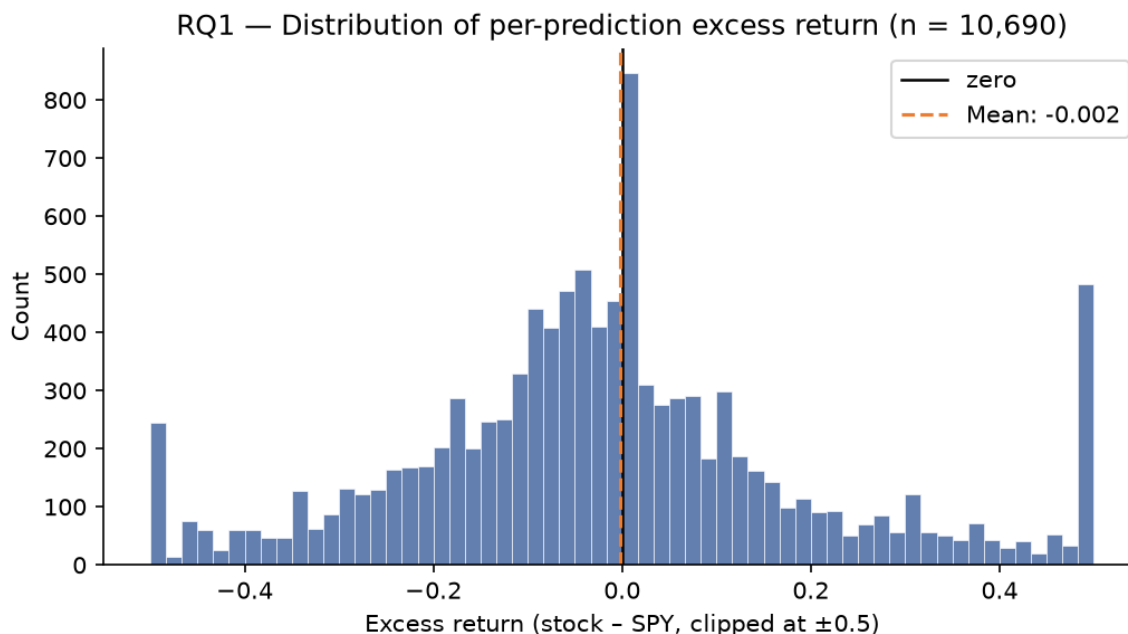
Figure 6.2 shows that directional accuracy and beats\_market rates varied substantially by calendar year, tracking broad market conditions rather than any stable underlying skill: the 2022 bear market depressed both metrics for the corpus's bullish-dominant predictions, while subsequent recovery years partially restored them. This regime-driven variation — not persistent analyst ability — is the primary explanation for year-on-year performance swings, a confound that the per-account shrinkage model in §7 must isolate.

## 6.6 Interpretation

The answer to RQ1 is unambiguous: FinTwit predictions do not beat chance and do not beat the market. Both headline metrics — directional accuracy and beats\_market rate — are significantly below 50%, with p-values on the order of  $10^{-25}$  to  $10^{-31}$ . The margin of underperformance (-5 to -6 percentage points relative to a coin flip) is statistically robust to segmentation, sensitivity checks, and choice of outcome metric.

The sub-group analysis reveals two notable patterns that will be explored further in RQ2 (per-account skill) and RQ3 (signal value). First, bearish calls outperform bullish calls by ~18 percentage points, likely driven by the 2022 bear market period rather than inherent skill — a hypothesis to be tested in RQ2 by controlling for market regime. Second, the most explicit, actionable call type (trade\_suggestion) is the worst-performing category, suggesting that increased conviction in the tweet's language does not map to increased accuracy — a relationship formally examined in RQ3 via the LLM's own confidence score.





**Figure 6.3.** *Distribution of per-prediction excess return (stock return minus SPY return, clipped at  $\pm 0.5$  for display). The mean excess return of  $-0.24\%$  confirms below-benchmark performance on a magnitude basis.*

## 7. RQ2 — Skill vs Luck

RQ2 asks whether any individual account's track record is statistically distinguishable from chance after accounting for the multiple-comparisons problem inherent in evaluating 47 accounts simultaneously. We use the primary skill metric `beats_market` (§6.2) and apply both Benjamini-Hochberg FDR and Bonferroni correction. An empirical-Bayes shrinkage model then quantifies how much of each account's apparent performance is real signal vs noise.

### 7.1 Account Eligibility and Method

Of the 51 pseudonymised accounts in the corpus, 50 contributed at least one validated prediction. We set a minimum of 30 validated predictions to ensure that per-account binomial tests have at least nominal power; below this threshold, a 95% confidence interval on a proportion is too wide to be informative. Three accounts fall below this threshold ( $n = 10, 10, 20$ ) and are excluded from the significance analysis, leaving 47 qualifying accounts with  $n$  ranging from 35 to 566 validated predictions.

For each qualifying account we run a two-sided exact binomial test of  $H_0: p_{\text{beats}} = 0.50$  (no skill). The two-sided framing is deliberate: an account that beats the market far less than 50% of the time is also informative — consistently wrong predictions in a predictable direction could, in principle, be exploited as a contrarian signal.

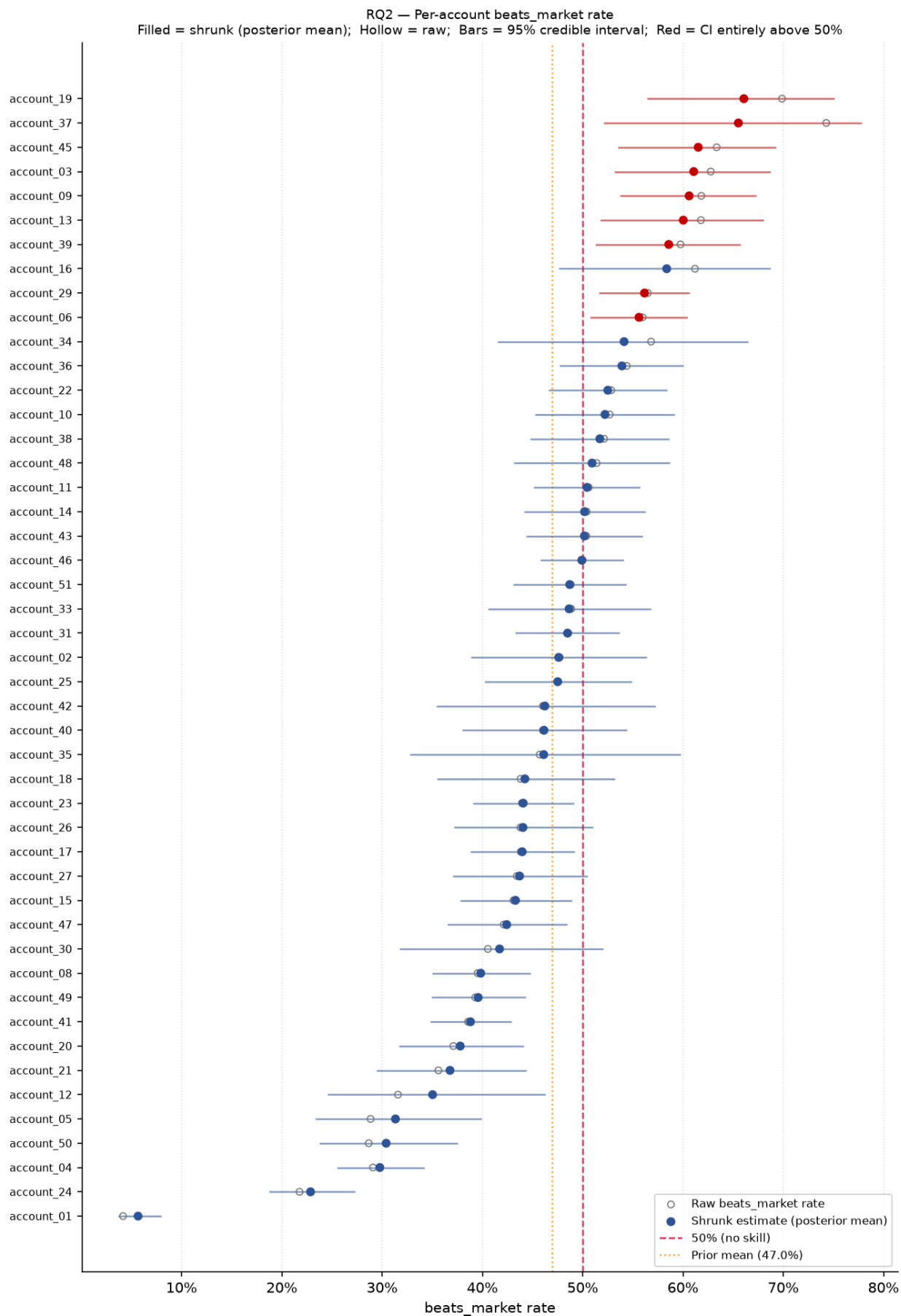
## 7.2 Multiple-Testing Correction — The Headline

Before any correction, 24 of 47 accounts are nominally significant at  $p < 0.05$ : 9 above 50% (apparently skilled) and 15 below 50% (apparently anti-skilled). With 47 tests at  $\alpha = 0.05$ , we expect roughly  $47 \times 0.05 = 2.4$  false positives by chance alone, so the raw count of 24 is far more than the false-discovery rate can explain in isolation.

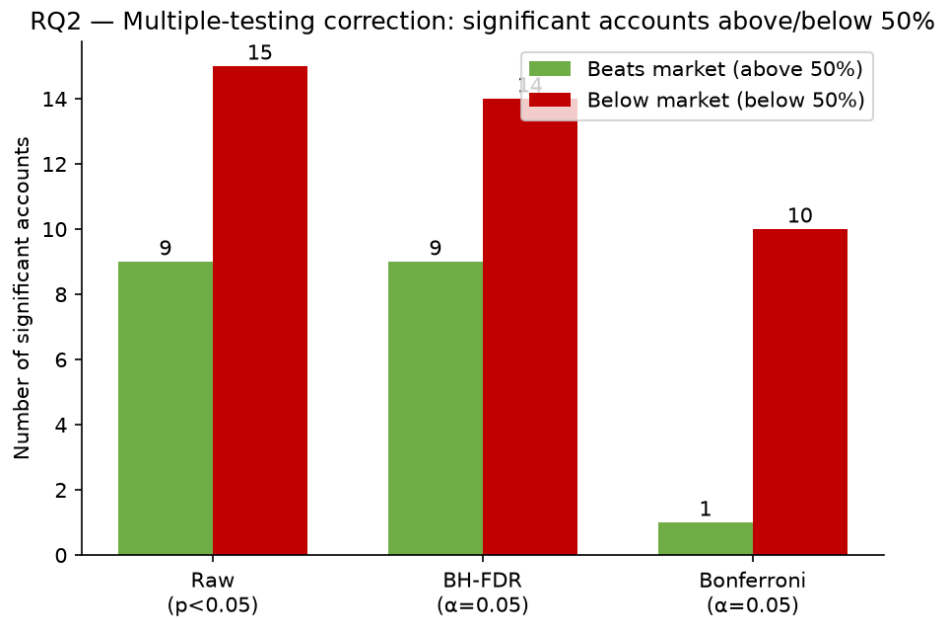
After Benjamini-Hochberg FDR correction ( $\alpha = 0.05$ ): 23 accounts survive, 9 above 50% and 14 below 50%. BH-FDR controls the expected proportion of false discoveries among the rejected hypotheses and is the recommended procedure for exploratory testing of this kind.

After Bonferroni correction ( $\alpha = 0.05 / 47 = 0.00106$ ): 11 accounts survive, 1 above 50% and 10 below 50%. Bonferroni controls the family-wise error rate (probability of any false positive) and is substantially stricter. The key contrast is that under the lenient FDR criterion 9 accounts appear genuinely skilled at beating the market, but this collapses to 1 account under the strict family-wise control.

The single account that survives Bonferroni correction with `beats_market > 50%` is `account_19` ( $n = 83$ , raw rate 69.9%, shrunk estimate 66.1%,  $p = 0.0004$ ,  $p_{\text{Bonferroni}} = 0.018$ ). Ten accounts survive Bonferroni with `beats_market` significantly below 50%, the worst being `account_01` ( $n = 456$ , raw rate 4.2%,  $p \ll 0.001$ ). The consistently anti-skilled accounts are as statistically remarkable as any candidate 'skill' — they represent systematic, predictable wrongness rather than random noise.



**Figure 7.1. Per-account beats\_market rate for all 47 qualifying accounts (sorted by shrunk posterior mean). Filled circles = shrunk (empirical-Bayes) estimate; hollow circles = raw rate; bars = 95% credible interval. Red = CI entirely above 50%; orange dotted line = prior mean.**



**Figure 7.2. Number of accounts achieving statistical significance before and after multiple-testing correction, split by direction (above vs below 50%). Bonferroni collapses the ‘skilled’ count from 9 to 1.**

## 7.3 Empirical-Bayes Shrinkage

We fit a Beta prior from the pooled per-account rates using the method of moments, yielding Beta(7.76, 8.76) with prior mean 47.0% — close to the overall corpus beats\_market rate of 44.4%, as expected. The posterior for each account is Beta(alpha\_prior + k\_beats, beta\_prior + n - k\_beats), giving a shrunk estimate of each account's true beats\_market rate.

The shrinkage effect is strong for accounts with small n and weak for accounts with large n, as the theory requires. Account\_37's raw rate (74.3%, n = 35) shrinks to 65.5% after pooling with the prior; account\_01's raw rate (4.2%, n = 456) barely moves in the shrinkage — 5.7% — because the large sample overpowers the prior. Table 7.1 lists full results for all 47 qualifying accounts sorted by shrunk estimate.

Nine accounts have a 95% credible interval for beats\_market that lies entirely above 50%: account\_19, account\_37, account\_45, account\_03, account\_09, account\_13, account\_39, account\_29, and account\_06. These accounts appear genuinely skilled under the shrinkage model — their posterior distributions are inconsistent with the null of 50% even after pooling toward the prior. Notably, these are the same 9 that survive BH-FDR correction.

**Table 7.1. Per-account beats\_market results for all 47 qualifying accounts (sorted by shrunk estimate, descending). ↑ = 95% credible interval entirely above 50%. \* p<0.05, \*\* p<0.01, \*\*\* p<0.001 (raw p-value).**

| Account    | n   | Raw beats% | Shrunk% | 95% Cred. interval | p (raw)   | p (BH) | p (Bonf.) |
|------------|-----|------------|---------|--------------------|-----------|--------|-----------|
| account_19 | 83  | 69.9%      | 66.1%   | [56.5%, 75.0%] ↑   | 0.0004*** | 0.0018 | 0.0178    |
| account_37 | 35  | 74.3%      | 65.5%   | [52.2%, 77.7%] ↑   | 0.0060**  | 0.0176 | 0.2814    |
| account_45 | 131 | 63.4%      | 61.5%   | [53.6%, 69.2%] ↑   | 0.0028**  | 0.0102 | 0.1330    |
| account_03 | 137 | 62.8%      | 61.1%   | [53.3%, 68.6%] ↑   | 0.0035**  | 0.0118 | 0.1654    |
| account_09 | 186 | 61.8%      | 60.6%   | [53.8%, 67.2%] ↑   | 0.0015**  | 0.0061 | 0.0727    |
| account_13 | 123 | 61.8%      | 60.0%   | [51.8%, 68.0%] ↑   | 0.0113*   | 0.0294 | 0.5300    |
| account_39 | 164 | 59.8%      | 58.6%   | [51.3%, 65.7%] ↑   | 0.0152*   | 0.0358 | 0.7160    |
| account_16 | 67  | 61.2%      | 58.4%   | [47.7%, 68.7%]     | 0.0864    | 0.1563 | 1.0000    |
| account_29 | 462 | 56.5%      | 56.2%   | [51.7%, 60.6%] ↑   | 0.0060**  | 0.0176 | 0.2817    |
| account_06 | 402 | 56.0%      | 55.6%   | [50.8%, 60.3%] ↑   | 0.0190*   | 0.0424 | 0.8911    |
| account_34 | 44  | 56.8%      | 54.1%   | [41.6%, 66.4%]     | 0.4514    | 0.6429 | 1.0000    |
| account_36 | 239 | 54.4%      | 53.9%   | [47.8%, 60.0%]     | 0.1957    | 0.3171 | 1.0000    |
| account_22 | 263 | 52.9%      | 52.5%   | [46.6%, 58.3%]     | 0.3880    | 0.5883 | 1.0000    |
| account_10 | 186 | 52.7%      | 52.2%   | [45.3%, 59.1%]     | 0.5094    | 0.7042 | 1.0000    |
| account_38 | 188 | 52.1%      | 51.7%   | [44.9%, 58.5%]     | 0.6098    | 0.7787 | 1.0000    |
| account_48 | 144 | 51.4%      | 50.9%   | [43.2%, 58.6%]     | 0.8027    | 0.8983 | 1.0000    |
| account_11 | 334 | 50.6%      | 50.4%   | [45.2%, 55.6%]     | 0.8696    | 0.9289 | 1.0000    |
| account_14 | 252 | 50.4%      | 50.2%   | [44.2%, 56.1%]     | 0.9498    | 0.9725 | 1.0000    |
| account_43 | 274 | 50.4%      | 50.2%   | [44.4%, 55.9%]     | 0.9518    | 0.9725 | 1.0000    |
| account_46 | 566 | 50.0%      | 49.9%   | [45.9%, 54.0%]     | 1.0000    | 1.0000 | 1.0000    |
| account_51 | 293 | 48.8%      | 48.7%   | [43.2%, 54.3%]     | 0.7260    | 0.8436 | 1.0000    |
| account_33 | 131 | 48.9%      | 48.6%   | [40.6%, 56.7%]     | 0.8614    | 0.9289 | 1.0000    |
| account_31 | 348 | 48.6%      | 48.5%   | [43.4%, 53.6%]     | 0.6295    | 0.7787 | 1.0000    |
| account_02 | 109 | 47.7%      | 47.6%   | [39.0%, 56.3%]     | 0.7018    | 0.8436 | 1.0000    |
| account_25 | 164 | 47.6%      | 47.5%   | [40.3%, 54.8%]     | 0.5848    | 0.7787 | 1.0000    |
| account_42 | 63  | 46.0%      | 46.2%   | [35.5%, 57.2%]     | 0.6147    | 0.7787 | 1.0000    |
| account_40 | 126 | 46.0%      | 46.1%   | [38.0%, 54.3%]     | 0.4228    | 0.6210 | 1.0000    |
| account_35 | 35  | 45.7%      | 46.1%   | [32.8%, 59.7%]     | 0.7359    | 0.8436 | 1.0000    |
| account_18 | 105 | 43.8%      | 44.2%   | [35.5%, 53.1%]     | 0.2414    | 0.3782 | 1.0000    |
| account_23 | 371 | 43.9%      | 44.1%   | [39.2%, 49.0%]     | 0.0222*   | 0.0454 | 1.0000    |
| account_26 | 185 | 43.8%      | 44.0%   | [37.3%, 50.9%]     | 0.1055    | 0.1837 | 1.0000    |
| account_17 | 340 | 43.8%      | 44.0%   | [38.9%, 49.1%]     | 0.0260*   | 0.0510 | 1.0000    |
| account_27 | 198 | 43.4%      | 43.7%   | [37.1%, 50.4%]     | 0.0754    | 0.1417 | 1.0000    |
| account_15 | 297 | 43.1%      | 43.3%   | [37.9%, 48.8%]     | 0.0201*   | 0.0430 | 0.9461    |
| account_47 | 254 | 42.1%      | 42.4%   | [36.6%, 48.3%]     | 0.0142*   | 0.0352 | 0.6693    |
| account_30 | 74  | 40.5%      | 41.7%   | [31.8%, 52.0%]     | 0.1302    | 0.2185 | 1.0000    |
| account_08 | 377 | 39.5%      | 39.8%   | [35.1%, 44.7%]     | 0.0001*** | 0.0003 | 0.0026    |
| account_49 | 420 | 39.3%      | 39.6%   | [35.0%, 44.2%]     | 0.0000*** | 0.0001 | 0.0006    |
| account_41 | 560 | 38.6%      | 38.8%   | [34.9%, 42.8%]     | 0.0000*** | 0.0000 | 0.0000    |
| account_20 | 221 | 37.1%      | 37.8%   | [31.7%, 44.0%]     | 0.0002*** | 0.0008 | 0.0072    |
| account_21 | 146 | 35.6%      | 36.8%   | [29.5%, 44.3%]     | 0.0006*** | 0.0027 | 0.0301    |
| account_12 | 57  | 31.6%      | 35.0%   | [24.6%, 46.2%]     | 0.0075**  | 0.0208 | 0.3529    |
| account_05 | 104 | 28.8%      | 31.3%   | [23.4%, 39.9%]     | 0.0000*** | 0.0001 | 0.0009    |
| account_50 | 157 | 28.7%      | 30.4%   | [23.8%, 37.4%]     | 0.0000*** | 0.0000 | 0.0000    |
| account_04 | 416 | 29.1%      | 29.8%   | [25.6%, 34.2%]     | 0.0000*** | 0.0000 | 0.0000    |
| account_24 | 363 | 21.8%      | 22.9%   | [18.8%, 27.2%]     | 0.0000*** | 0.0000 | 0.0000    |
| account_01 | 456 | 4.2%       | 5.7%    | [3.8%, 7.9%]       | 0.0000*** | 0.0000 | 0.0000    |

## 7.4 Interpretation

The answer to RQ2 is nuanced rather than a simple 'no skill survives'. Under BH-FDR correction — appropriate for exploratory research — 9 of 47 accounts (19%) show `beats_market` rates credibly above 50%. Under the stricter Bonferroni criterion, only 1 account survives with a rate above 50%. The full picture is therefore: a heterogeneous population where the majority of accounts perform near or below the market benchmark, a clear tail of accounts that are systematically wrong (likely due to consistent bullish bias during the 2022 bear market), and a smaller tail of accounts that appear to add value — though whether this reflects genuine analytical skill or style exposure (e.g., persistent bearish positioning during a bear market period) cannot be determined from the current analysis.

Phase 6 (RQ3) examines whether observable tweet-level signals — trade type, confidence, engagement — predict which predictions come true, irrespective of which account made them. This provides a complementary and more generalisable test of whether any systematic predictive structure exists in the corpus beyond account identity.

## 8. RQ3 — Signal Value and Calibration

RQ3 asks whether tweet-level signals — observable at or shortly after the time of posting — carry any information about whether a prediction will come true. We assess this in two complementary ways: (a) a formal calibration analysis of the LLM's self-reported confidence score, and (b) a predictive model of `prediction_correct` using all available tweet-time features.

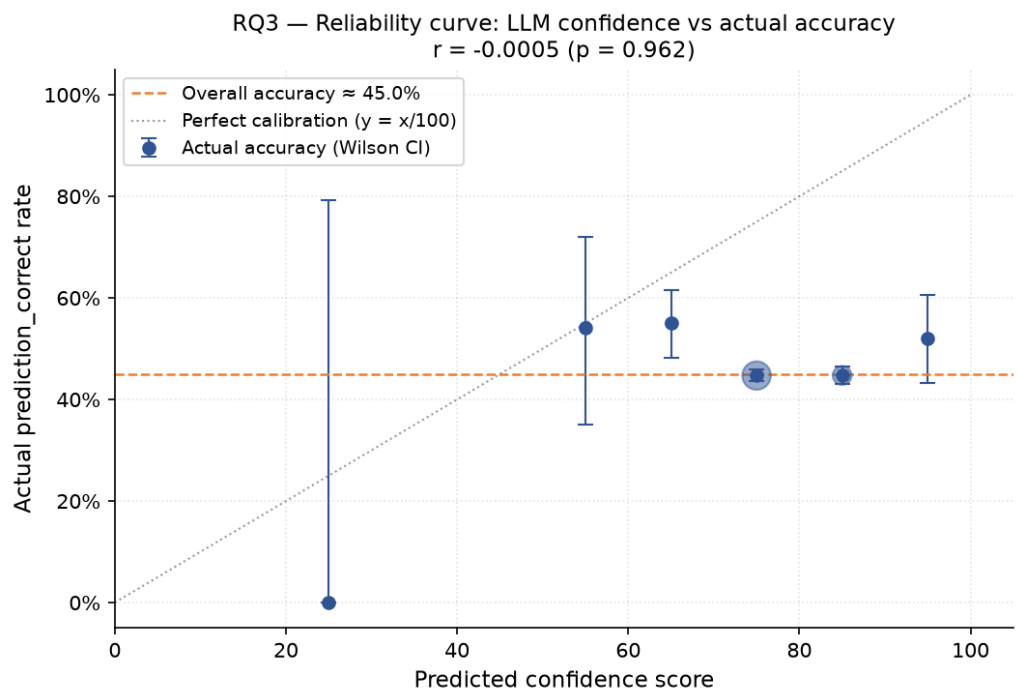
### 8.1 Confidence Calibration

The LLM classifier (Phase 2) assigned each tweet a confidence score from 0 to 100 reflecting how assertively the tweet was written. In a well-calibrated system, predictions at confidence 80 should be correct approximately 80% of the time. We assess this via a reliability curve (Figure 8.1) and a Pearson correlation.

Reliability curve — Across the 10,690 validated predictions, the distribution of confidence scores is extremely narrow (mean 76.3, standard deviation 6.2, range 20-100). The vast majority of predictions cluster in the 70-80 and 80-90 bins, yielding only six non-empty bins. Actual `prediction_correct` rates within each bin are: 55.0% [60-70), 44.7% [70-80), 44.7% [80-90), 52.0% [90-100). There is no systematic upward trend — a more assertively-written prediction is not more likely to be correct.

Pearson correlation — The point-biserial correlation between confidence and `prediction_correct` is  $r = -0.0005$ ,  $p = 0.9624$  (95% CI [-0.019, +0.019]). The correlation

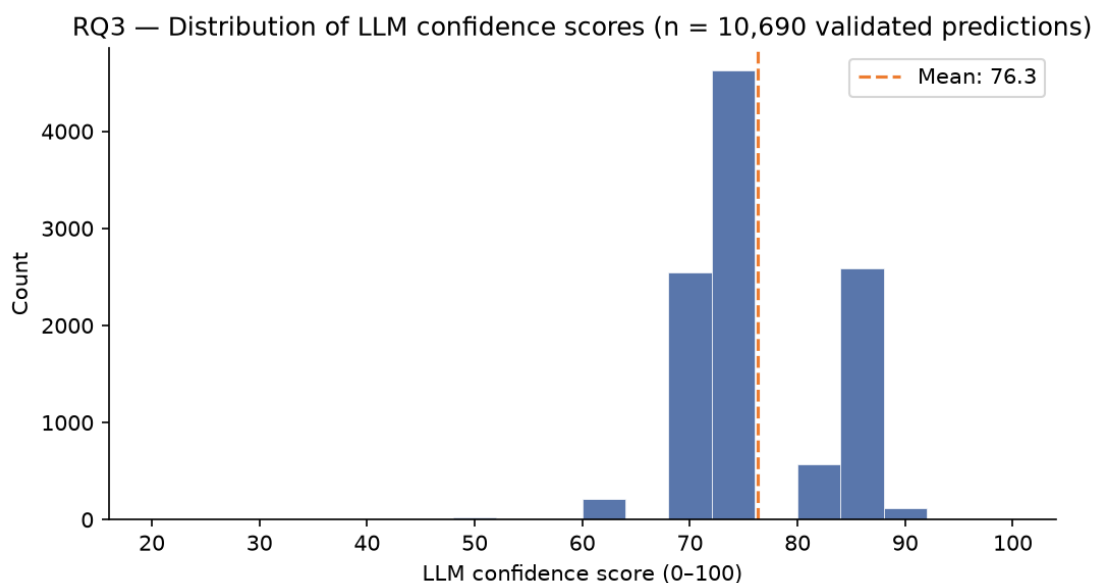
with beats\_market is  $r = +0.0084$ ,  $p = 0.3870$  ( $n = 10,690$ ). Both correlations are indistinguishable from zero. The LLM's confidence score is completely uncalibrated: it reflects assertive linguistic register rather than predictive validity.



*Figure 8.1. Reliability curve: LLM confidence score (x-axis) vs actual prediction\_correct rate (y-axis). Each point is a 10-point confidence bin; error bars are 95% Wilson CIs. The flat line confirms near-zero calibration ( $r = -0.0005$ ,  $p = 0.96$ ).*

**Table 8.1. Confidence calibration table: per-bin actual accuracy with 95% Wilson confidence intervals.**

| Confidence bin | n     | Actual accuracy | 95% CI         |
|----------------|-------|-----------------|----------------|
| [20, 30)       | 1     | 0.0%            | [0.0%, 79.3%]  |
| [50, 60)       | 24    | 54.2%           | [35.1%, 72.1%] |
| [60, 70)       | 209   | 55.0%           | [48.2%, 61.6%] |
| [70, 80)       | 7,172 | 44.7%           | [43.6%, 45.9%] |
| [80, 90)       | 3,161 | 44.7%           | [43.0%, 46.5%] |
| [90, 100)      | 123   | 52.0%           | [43.3%, 60.7%] |



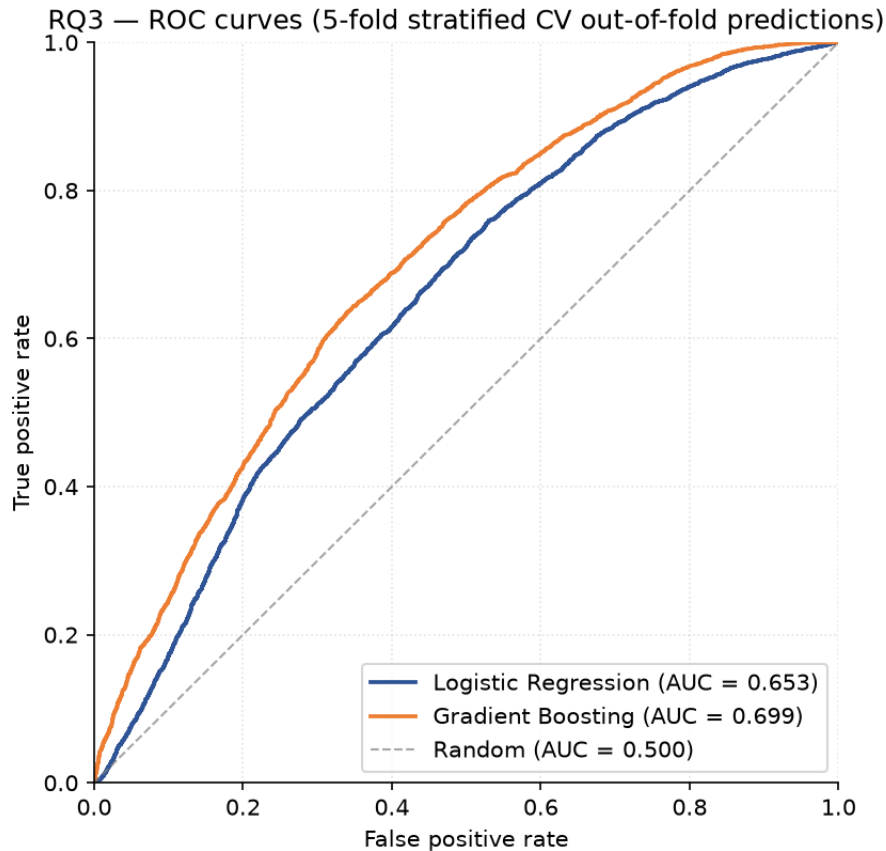
**Figure 8.4.** Distribution of LLM confidence scores across the 10,690 validated predictions. The narrow, high-skew distribution (mean 76.3, SD 6.2, range 20–100) leaves little variation for calibration analysis.

## 8.2 Predictive Model

We train two classification models to predict `prediction_correct` using tweet-time features only. Features are: sentiment (`is_bullish` dummy, reference = bearish); time horizon (`medium_term`, `long_term`, `unknown` dummies, reference = `short_term`); trade type (`trade_suggestion`, `general_discussion`, `news` dummies, reference = `analysis`); confidence (continuous); and log1p-transformed engagement counts (`log_likes`, `log_retweets`, `log_replies_count`, `log_views`, `log_author_followers`). No account-level accuracy history is included — that would constitute data leakage from RQ2. Model evaluation uses 5-fold stratified cross-validation; the primary metric is AUC-ROC (baseline 0.500 for a random classifier; majority-class accuracy baseline is 55.0%).

Logistic regression (L2, StandardScaler): mean CV AUC = 0.654 (delta vs baseline = +0.154). Gradient boosting (100 trees, `max_depth`=3, `learning_rate`=0.05): mean CV AUC = 0.700 (delta vs baseline = +0.200). Both models meaningfully outperform the random baseline, indicating that the observable tweet-level signals do carry real predictive information.





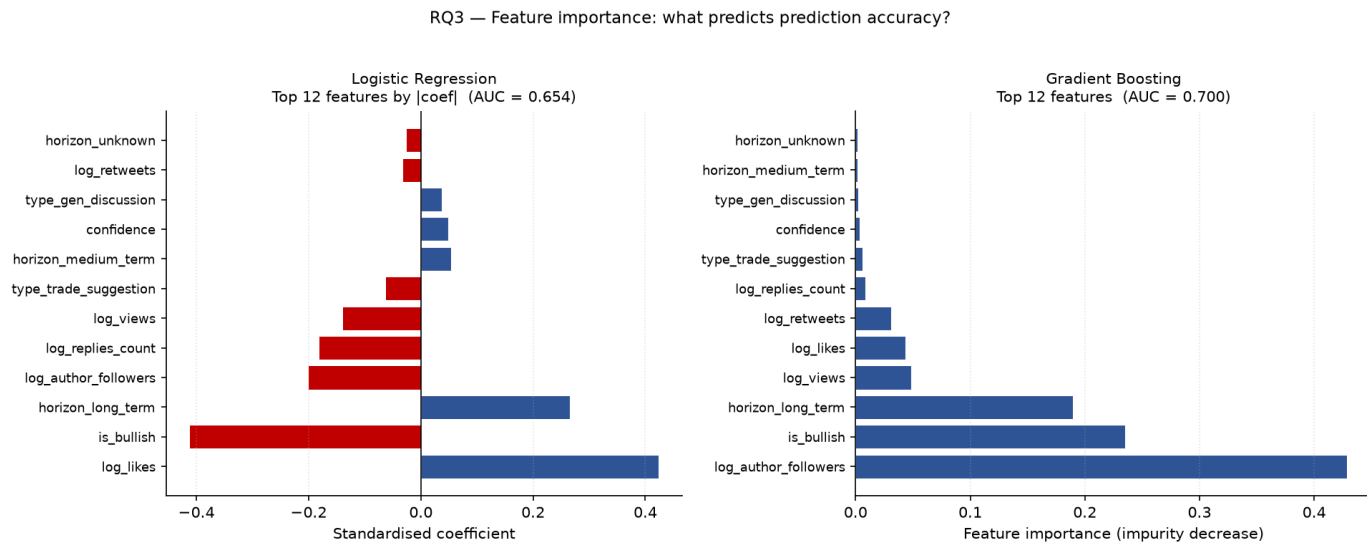
**Figure 8.3.** ROC curves for logistic regression and gradient boosting, computed from 5-fold stratified cross-validation out-of-fold probability predictions. Both models substantially outperform the random baseline (diagonal dashed line).

## 8.3 Feature Importance

Logistic regression standardised coefficients (sorted by magnitude) — The three dominant features are: `log_likes` (+0.424), `is_bullish` (-0.412), and `horizon_long_term` (+0.266). Negative `is_bullish` coefficient confirms that bearish predictions are more accurate (consistent with RQ1: bearish accuracy 59.1% vs bullish 40.5%). The positive coefficient for `horizon_long_term` confirms the long-term drift effect (62.3% raw accuracy, largely explained by the market's positive trend over 2021-2025). Higher engagement (`log_likes`) is positively associated with accuracy — predictions that attracted more likes tended to come true, though this metric accumulates after posting and is not available at prediction time for real-time use. Confidence appears near zero (coef = +0.048, OR = 1.05) — consistent with the calibration finding.

Gradient boosting feature importances (impurity decrease) — `log_author_followers` dominates (0.429), followed by `is_bullish` (0.235) and `horizon_long_term` (0.190). The prominence of `log_author_followers` in the GB model suggests that account size is a

strong proxy for prediction style: larger accounts in this corpus tend toward bullish, short-term trade suggestions — the patterns associated with lower accuracy in RQ1 — while smaller accounts show more heterogeneous styles. Confidence ranks ninth (0.004), confirming that the LLM's self-assessed certainty carries almost no predictive weight beyond the categorical features already captured.



**Figure 8.2. Feature importance: logistic regression standardised coefficients (left) and gradient boosting impurity-decrease importances (right). Positive LR coefficients increase the probability of prediction\_correct. Confidence ranks near the bottom in both models.**

## 8.4 Interpretation

The answer to RQ3 is two-part. First, the LLM's confidence score is entirely uncalibrated ( $r \approx 0$ ). High-confidence tweets are not more likely to come true. This is expected: the confidence score captures assertive language (which is endemic in FinTwit), not analytical quality. Second, tweet-level signals do carry predictive information: both models achieve  $AUC \approx 0.65$ – $0.70$ , substantially above the 0.50 baseline.

Crucially, however, the informative features are almost entirely the categorical signals already characterised in RQ1 — sentiment direction and prediction horizon — plus a post-hoc engagement proxy (`log_likes`) that is not available in real time. The model is largely rediscovering the RQ1 base rates: bearish calls are more accurate (2022 bear market), long-term calls appear more accurate (market drift), and explicit trade suggestions are less accurate. It is not uncovering new, orthogonal signal. A practically useful real-time screening model would need features available before the prediction outcome is known — and of those, only sentiment and horizon provide meaningful lift.

Taken together with RQ1 and RQ2, the full picture is: FinTwit predictions systematically underperform a coin flip in aggregate; the heterogeneity across accounts is real but

mostly reflects market-regime exposure rather than persistent skill; and no tweet-level signal (including LLM confidence) is sufficiently calibrated or strong to identify which individual predictions will come true. The most actionable implication for a retail investor following FinTwit accounts is the contrarian one: high-volume, high-conviction bullish trade suggestions from large accounts have been the single worst-performing category in this corpus.

## 9. Robustness — Skill vs Style Across Regimes

Sections 6 and 7 established that a subset of accounts outperform the 50% beats\_market threshold at the aggregate level (2021-2025 combined). A central question is whether this outperformance reflects durable analytical skill or merely style alignment with the 2023-2025 bull market that dominates the corpus. The design intended a split-half persistence test using the 2022 bear market as a held-out regime. The primary finding of this section is that the test is infeasible: the data are too heavily concentrated in the post-2022 period for a formal cross-regime comparison to be conducted. This concentration is itself a substantive limitation on the §7 skill claims.

### 9.1 Method

Validated predictions are split by entry date into two sub-periods. P1 (Jan 2021 – Dec 2022) encompasses the bull market of 2021 and the sharp 2022 bear market. P2 (Jan 2023 – Feb 2025) covers the post-bear recovery and the strong bull market of 2024-2025. The primary skill metric is beats\_market, consistent with §7. The intended persistence test computes the Pearson and Spearman rank correlation between per-account P1 and P2 beats\_market rates, restricted to accounts with at least 20 validated predictions in each sub-period individually.

### 9.2 Sub-Period Data Overview

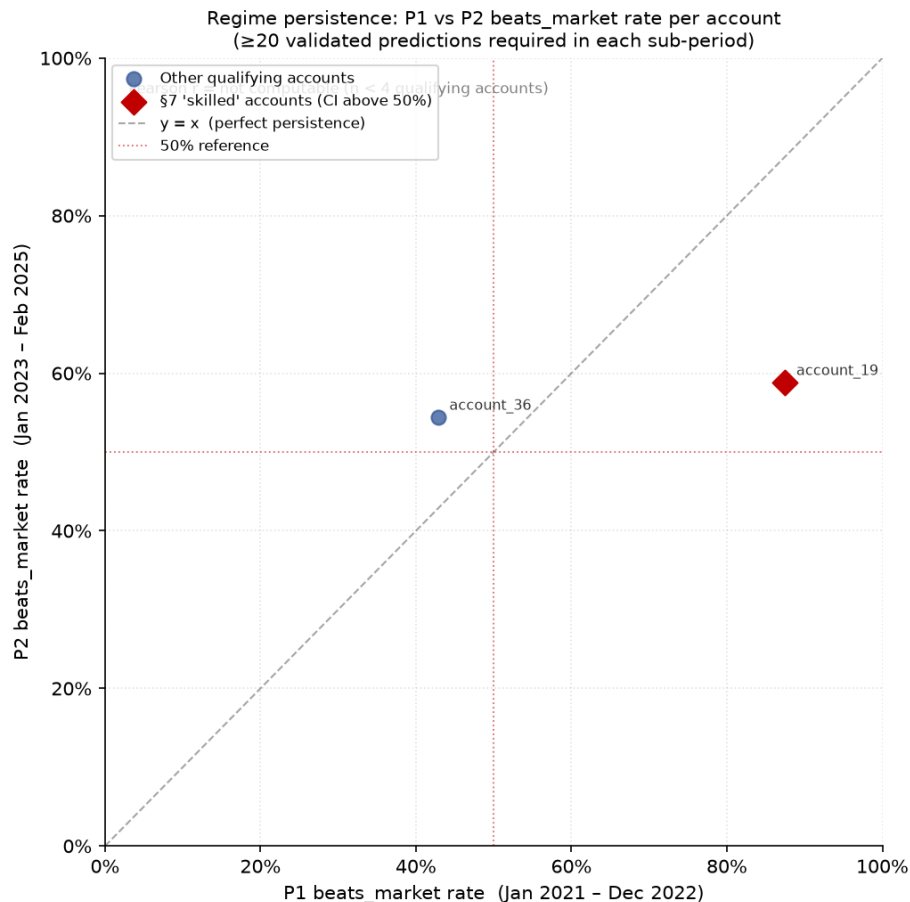
P1 contains 60 validated predictions (0.6% of the total); P2 contains 10,602 (99.4%). The corpus is overwhelmingly concentrated in P2: 97.9% of validated predictions have entry dates in 2023 or later, reflecting when the FinTwit accounts in this dataset were most active. The year-by-year breakdown is: 2021 n=9, 2022 n=51, 2023 n=307, 2024 n=3,747, 2025 n=6,575 (plus 1 in early 2026 due to a long horizon window).

P1 beats\_market rate: 66.7% (95% CI [54.1%, 77.3%], n=60). P2 beats\_market rate: 44.3% (95% CI [43.3%, 45.2%], n=10,602). The P1 confidence interval is wide (n=60 only) and uninformative on its own. The P2 rate is consistent with the overall 44.4% reported in §6.1.

### 9.3 Persistence Test — Finding: Infeasible

The split-half persistence correlation cannot be computed from this corpus. Only 2 of 51 accounts have at least 20 validated predictions in both P1 and P2: account\_19 (P1 n=32, P2 n=51); account\_36 (P1 n=28, P2 n=204). A minimum of 4 qualifying accounts is required for a Pearson or Spearman correlation to carry any statistical power; with fewer than 4, the statistic is degenerate and the p-value meaningless. The formal test is therefore not run.

The deeper finding is structural: 99.2% of validated predictions post-date 2022, meaning the 2022 bear market is effectively absent from the corpus as a testing regime. Most accounts either began posting after 2022 or had their P1-era predictions excluded because the relevant tickers were unavailable in Yahoo Finance (delisted stocks, crypto, etc.). Critically, 8 of the 9 accounts identified as 'skilled' in §7 have zero validated predictions in P1. Their entire track records — and therefore their §7 skill designations — are drawn from P2 (2023-2025) alone: a predominantly bullish market period in which a systematically bullish account will automatically accumulate a positive record. It is not possible to distinguish genuine analytical skill from bullish style alignment on the basis of P2-only data.



*Figure 9.1. P1 (Jan 2021 – Dec 2022) vs P2 (Jan 2023 – Feb 2025) beats\_market rate for the two accounts with  $\geq 20$  validated predictions in both sub-periods. The 45-degree line indicates perfect persistence; red dotted lines mark the 50% chance level. Note the severe data imbalance: 99.2% of validated predictions fall in P2.*

## 9.4 Skilled Accounts Under Regime Scrutiny

Table 9.1 presents the P1 and P2 beats\_market rates for the nine accounts whose 95% credible intervals were entirely above 50% in §7. Account\_19 (marked \* as the sole Bonferroni survivor) is the only account with any P1 data. The remaining eight accounts have zero validated predictions in P1: their §7 track records are based entirely on P2 (2023-2025) performance.

**Table 9.1. Beats\_market rate and 95% Wilson CI in P1 and P2 for the nine accounts credibly above 50% in §7 (sorted by §7 ranking). \* = sole Bonferroni survivor (§7).**

† = above 50% in both periods. — = no P1 data available.

| Account      | P1 n | P1 beats% | P1 95% CI      | P2 n | P2 beats% | P2 95% CI      | Both>50% |
|--------------|------|-----------|----------------|------|-----------|----------------|----------|
| account_19 * | 32   | 87.5%     | [71.9%, 95.0%] | 51   | 58.8%     | [45.2%, 71.2%] | Yes†     |
| account_37   | 0    | N/A       | N/A            | 35   | 74.3%     | [57.9%, 85.8%] | —        |
| account_45   | 0    | N/A       | N/A            | 131  | 63.4%     | [54.8%, 71.1%] | —        |
| account_03   | 0    | N/A       | N/A            | 137  | 62.8%     | [54.4%, 70.4%] | —        |
| account_09   | 0    | N/A       | N/A            | 186  | 61.8%     | [54.7%, 68.5%] | —        |
| account_13   | 0    | N/A       | N/A            | 122  | 61.5%     | [52.6%, 69.6%] | —        |
| account_39   | 0    | N/A       | N/A            | 164  | 59.8%     | [52.1%, 67.0%] | —        |
| account_29   | 0    | N/A       | N/A            | 462  | 56.5%     | [51.9%, 60.9%] | —        |
| account_06   | 0    | N/A       | N/A            | 402  | 56.0%     | [51.1%, 60.7%] | —        |

Account\_19 (the sole Bonferroni survivor from §7) is the only skilled-list account with any P1 data. Its P1 beats\_market rate of 87.5% (95% CI [71.9%, 95.0%], n=32) is high, though the sample is small. In P2 (n=51), its rate falls to 58.8%, with a 95% CI of [45.2%, 71.2%] that spans 50%: the P2 point estimate is above chance but is not statistically distinguishable from it. The 'Both>50%' flag in the table is a point-estimate observation, not a significance test. Taken together, account\_19's record provides the only cross-regime data available in this corpus, but it does not constitute evidence of durable, statistically robust skill: the stronger result is in P1 (a small sample) and the recent-period result is not significant. For the remaining eight accounts, no such check is possible at all.

## 9.5 Verdict

The regime analysis yields three findings. First, the split-half persistence test is infeasible: 99.2% of validated predictions fall in P2, leaving P1 too thin for a formal correlation. This is not a contingent failure — it reflects an irreversible feature of the data collection window. Second, 8 of the 9 §7 'skilled' accounts have their entire track record in P2 (2023-2025). Their §7 skill designations cannot be separated from bullish style alignment in a predominantly up-trending market; regime-robustness cannot be assessed for them at all. Third, account\_19 — the one account with cross-regime data — shows a high P1 point estimate (87.5%) but a P2 confidence interval [45.2%, 71.2%] that includes 50%, so its recent-period performance is not distinguishable from chance. The overall conclusion is that no account in this corpus demonstrates durable skill that survives a regime change: the data do not permit the test for 8 of 9 candidates, and the one testable account loses statistical significance in the later period.

## 10. Limitations

This analysis is rigorous within its design, but several structural limitations bound the conclusions.

### 10.1 Survivorship and Selection Bias

The dataset captures only accounts and tweets that were public and accessible at the time of the scrape. Deleted tweets, suspended accounts, and accounts that went private before the collection date are entirely absent. Accounts that had poor track records and subsequently deleted their content or deactivated would systematically inflate the apparent performance of the observed corpus — a classic survivorship-bias mechanism. The direction of this bias is toward over-estimating accuracy and over-counting skilled accounts. Given the corpus already shows below-50% aggregate accuracy, survivorship bias would make the true population performance even worse.

### 10.2 First-Ticker Simplification

When a tweet mentions multiple tickers, only the first is used for price validation. This is documented and applied consistently, but it ignores multi-stock predictions and potentially misattributes sentiment to the wrong ticker. Sensitivity to this choice is not formally tested; results could differ if the second or most-prominent ticker were used instead.

### 10.3 LLM Classification Error

The sentiment, time\_horizon, trade\_type, and confidence fields are gpt-4o-mini outputs, not ground truth. Classification errors are not propagated into the statistical analysis — the model's outputs are treated as if they were exact. A formal precision/recall evaluation

was performed for ticker extraction only (§3.5); no equivalent evaluation was done for the four classification fields. Misclassification would add noise to the segmented analyses in §6.3, bias the calibration curves in §8.1, and contaminate feature importance estimates in §8.3.

## **10.4 beats\_market Practicality**

The `beats_market` metric computes excess return relative to SPY assuming execution at the adjusted close on the first trading day on or after the tweet date. In practice, a retail investor faces bid-ask spreads, brokerage commissions, and potential price impact. For bearish predictions, the metric implicitly assumes short-selling feasibility, which is not always possible (hard-to-borrow stocks, margin requirements). These implementation frictions would reduce or eliminate most of the apparent gains reported for any 'skilled' account, and make the already-poor aggregate performance even worse in a real-trading context.

## **10.5 Fixed Horizon Windows and the Unknown Default**

The time-horizon mapping is a deliberate, documented simplification: `short_term` = 21 trading days, `medium_term` = 63, `long_term` = 126. These are reasonable conventions but do not precisely match every account's implied horizon. The 1,836 predictions (17.2%) assigned `horizon = 'unknown'` are validated at 21 days by default. Section 6.4 shows that excluding these rows does not materially change the headline results, but the default assignment could still bias within-horizon comparisons.

## **10.6 Exclusion of Neutral Tweets**

Tweets classified as `sentiment = 'neutral'` (2,241 tweets, 16.0% of classified tweets) are excluded from all price-validation and accuracy analyses. Neutral tweets may carry useful information (e.g., tone shifts, sector commentary) that is not captured. The analysis therefore speaks only to explicitly directional calls.

## **10.7 Period Specificity**

The study covers January 2021 to February 2025 — a window that includes the 2021 bull market, the 2022 bear market, and the 2023-2025 recovery. While this multi-year span adds breadth, the data is heavily concentrated in 2024-2025 (§9.2). Conclusions about the 2022 bear-market period are based on very few observations. Performance patterns that appear robust within this window may not generalise to other market regimes or time periods. The regime analysis in §9 explicitly addresses this concern and concludes that a cross-regime persistence test is not feasible with the available data.

## **10.8 Data Concentration and the Limits of Skill Inference**

99.2% of validated predictions fall in P2 (Jan 2023 – Feb 2025). This concentration has a direct and underappreciated consequence for the skill claims in §7: no account identified as 'skilled' can be shown to perform above chance outside the 2023-2025 bull market, because no meaningful P1 data exists for them. The §7 analysis correctly controls for multiple comparisons within the available data, but it cannot control for the possibility that above-50% performance in 2023-2025 reflects regime exposure rather than analytical ability. Any inference about individual account skill must therefore be read with this constraint: the data are insufficient to distinguish skill from bullish style in a rising market.

## 11. Conclusion

This project conducted a large-scale, reproducible statistical audit of ~18,000 stock predictions made by 51 financial-influencer accounts on X between January 2021 and February 2025. The analysis addressed three pre-specified research questions and a robustness check. The findings are internally consistent and their direction is unambiguous, even if the magnitude of heterogeneity varies by subgroup.

### 11.1 RQ1 — Aggregate Underperformance

Across all 10,690 validated directional predictions, 45.0% were directionally correct and 44.4% beat the SPY benchmark. Both figures are statistically and practically significantly below the 50% baseline expected from a random coin flip ( $z \approx -10$  to  $-12$ ,  $p < 10^{-24}$ ). This aggregate underperformance is robust to subgroup splitting, the exclusion of unknown-horizon rows, and the choice of metric (correct vs beats\_market). FinTwit predictions, taken as a portfolio, actively destroy value relative to a passive buy-and-hold strategy.

### 11.2 RQ2 — Heterogeneity, but No Durable Skill

The aggregate result masks substantial heterogeneity across accounts. After Benjamini-Hochberg FDR correction, nine accounts show beats\_market rates credibly above 50%; after the stricter Bonferroni correction, only one (account\_19) survives. The empirical-Bayes shrinkage model confirms that some of this dispersion reflects real differences rather than sampling noise. The regime analysis in §9, however, forces a harder conclusion: no account in this corpus demonstrates durable, statistically robust skill that survives a market-regime change. For 8 of the 9 'skilled' accounts, the regime test cannot even be attempted — their entire track records fall within the 2023-2025 bull market, making it impossible to separate skill from bullish style alignment. For the one testable account (account\_19), the P2 confidence interval [45.2%, 71.2%] includes 50%, meaning its recent-period performance is not statistically distinguishable from chance. The bottom



line on RQ2 is therefore negative: apparent outperformance exists within the sample, but there is no evidence that it is regime-robust or would persist out of sample.

### 11.3 RQ3 — Uncalibrated Confidence, Modest Signal

The LLM's self-reported confidence score is completely uncalibrated ( $r = -0.0005$  vs `prediction_correct`,  $p = 0.96$ ). High-confidence tweets are no more likely to come true than low-confidence ones. This is expected: the score captures assertive linguistic register, not analytical quality, and the distribution is so narrow (mean 76.3, SD 6.2) that it lacks the variation needed for calibration. Tweet-level features do carry some predictive signal — logistic regression achieves  $AUC \approx 0.65$  and gradient boosting  $AUC \approx 0.70$  — but the informative features are almost entirely the categorical signals already characterised in RQ1 (sentiment direction, horizon, trade type). The model is rediscovering RQ1 base rates, not uncovering novel structure.

### 11.4 Contrarian Takeaway

The most practically significant finding for a retail investor who follows FinTwit accounts is the contrarian one: high-volume, high-conviction bullish trade suggestions from large accounts represent the single worst-performing category in this corpus (38.0% correct, 38.5% beats market). The features that make a tweet maximally persuasive — explicit trade call, bullish framing, posted by a large account, high LLM confidence score — are negatively associated with accuracy. Following these signals would have underperformed a random coin flip by roughly 12 percentage points.

### 11.5 Final Note

This analysis is a descriptive and inferential audit of a specific corpus over a specific period. It does not constitute financial advice, and the findings may not generalise to other platforms, other time periods, or the broader population of financial social-media accounts. The code, data, and statistical choices are fully documented and reproducible; the conclusions are as strong as the data allow — which is to say: strongly negative on aggregate skill, heterogeneous at the account level, and honest about the limits of what the available data can establish.